

EE 660 Power Management IC Design, 2025-26 Semester-II

Project 2, Date 25.02.2026 Weightage: 30%

Project Title: Design of a closed-loop voltage-mode buck converter in SCL 180nm CMOS Technology

Part 1: Circuit design using ideal components	8 th March	5 pts
Part 2: Transistor level design of supporting circuit block	8 th March	25 pts
Part 3: Integration and optimization	Will be updated	20 pts
Demo + Viva	Will be updated	10 pts

Reports for each part should be submitted on Google Classroom by the due date.

Completion of all preceding parts is mandatory.

Any subsequent parts shall not be assessed unless prior parts are completed.

Specifications:

Input Voltage (V_{in}): 3.3V

Output Voltage (V_{out}): 1.8V

Switching Frequency: 1 MHz

Load current: 100mA (light-load), 800mA (full-load), 500 mA (nominal-load)

Steady-state output voltage ripple: < 5 mV

Inductor: $L < 8$ μ H with $R_{dc,r}$ of 50 mOhm

Output capacitor: $C < 6$ μ F with R_{esr} of 20 mOhm

DC steady-state offset error: < 0.2 %

Load Regulation: $\leq 5\%$ variation from light-load to full-load

Line Regulation: $\leq 5\%$ for $\pm 10\%$ variation in input voltage

Transient Response: ≤ 20 mV overshoot/undershoot for light-load to full-load and vice-versa transition in 10ns (edge-time) with settling time (output voltage within $\pm 2\%$ of the regulated value) less than 125 ns.

Efficiency: $> 92\%$ over the entire load range

Design to match performance specs at 27C (nominal), 60C, and 100C

Integrate soft-start control circuit to limit peak inductor current to 1 A [Bonus points]

Note:

1. All achieved specifications (other than transient response and Load Regulation) should be reported at nominal load current
2. Overdrive voltage of all transistors should be > 0.15 V (nominally)
3. Only two ideal voltage sources at 1.8V and 3.3V can be used.

4. For efficiency measurement, must use “conservative” setting under transient analysis, and simulate at least 20 cycles after steady-state is achieved for accuracy.
5. Enable “skipDC” while doing the full-switching-circuit simulation.
6. For large circuits, saving all node voltages (by default) will take up storage and slow the simulation. Save only important nodes by select output->Save All-> check “selected”

Report:

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage) table of all transistors
- DC steady-state voltage error and steady-state voltage ripple values across the load current range (provide plots/tables as appropriate).
- Different losses like conduction loss, gate switching loss, dead-time switching loss, etc. separately in tabular format at 100mA, 500mA, 800mA.
- Step-by-step design methodology/derivations/calculations (in brief).
- Transient response plots should clearly mark the undershoot/overshoot voltage and settling time values, use proper simulation setting to have sufficient data points as per the time scale.
- Overall power consumption and its block-wise breakup should be provided.
- Plot efficiency across load range (multiple data points between light-load to full-load)
- Observation of impact of size of the PMOS/NMOS switches on factors like output and efficiency and explanation.
- Show switches gate drive waveforms and mark the dead-time (non-overlap time) and report it as a percentage of $1/f_{sw}$.
- Plots should be neatly labelled, and labels should be clearly visible.
- All Final achieved specification results should be provided in tabular format.
- Any other information as deemed necessary.

Suggested design flow:

- First, design and test comparator separately with $t_{delay} < 10\text{ns}$ (1% of T_{sw}) (you can use a ramp voltage as input to comparator against a reference voltage for testing). Then, design the ramp/clock generator (Recall that you will require comparator within ramp generator circuit as well) and integrate them to complete the PWM modulator.
- Next, integrate the gate driver, logic stages, switching stage, and LC filter to reach the design check point.
- Parallely, design OPAMP (feel free to borrow from previous designs) with $> 60\text{ dB}$ gain and $\sim 100\text{ kHz}$ UGB and integrate with Part 1 B circuit (replacing the ideal ‘vcvs’ element), close the loop via feedback network, and check load-transient response (only way to test stability!), adjust compensation if required.
- Integrate all elements together to complete your design!

Part 1: Circuit design using ideal components (5 pts)

Due on March 08, 2026, 11:59 PM

Digital submission is preferred over scanned handwritten submission

Part 1 A : Power stage design

A buck converter with $V_{in}=V_{dd}=3.3V$; $V_{out}=1.8V$; $f_{sw}=1MHz$ has load varying between 100mA (light-load) to 800mA (heavy-load), should have voltage ripple (peak-to-peak) $<10mV$ and should always operate in CCM. Minimum values of L and C have been calculated in assignment 2.

For the above buck converter, choose the L and C values of your liking (reasonably rounded value) higher than the minimum required (suggested to keep chosen LC product 2x to 3x times the minimum LC value). Simulate open-loop buck converter using ideal components (refer Appendix section below). Take a resistance of 50mohm in series with the inductor and ESR of 20 mohms in series with the Capacitor.

- What is the duty cycle that you set to get desired $V_{out}=1.8V$ under light-load and that under heavy-load? Is it different than the calculated values? Provide the reasoning.
- Under both light-load and heavy-load, provide steady-state waveforms for 1) output voltage (Mark and state the peak-to-peak voltage ripple) 2) switching node voltage, and 3) inductor current (Mark and state the peak-to-peak current ripple)
- Provide (zoomed-in) waveform at the transitions and mark the value of peak shoot-through current.

Part 1 B : Small signal compensator design

- For the buck converter simulated in Part 1 A ($V_{in}=V_{dd}=3.3V$; $V_{out}=1.8V$, $100mA < I_{load} < 800mA$, CCM, $f_{sw}=1MHz$, $ESR=20mohm$, $L_{dcr}=50mohm$), with the same chosen L and C values, additionally take $V_{ref}=1.2V$ and ramp voltage $V_H=2V$.
 - What is calculated LC double-pole frequency?
 - Simulate the loop transfer function i.e., AC Analysis (*refer Hint section below*) and report your circuit schematic,
 - For both light load and heavy load:** report 1) magnitude and phase bode plots (mark LC double-pole), 3) phase margin and gain margin, 4) comment on stability
- Add Type I compensation (*you have two options, either to use opamp or OTA, refer Hint sections below*).
 - What is the chosen value of C1 for $PM > 60$ degrees and $GM > 10dB$ (*Recall that high value of C1 reduces UGB, which is undesirable, try to maximize UGB*)
 - Report the numerical values of GM, PM, and UGB for both light-load and heavy-load conditions (Provide values in table format) and also provide the bode plots with markers at GM and PM crossover frequencies
 - Why are GM and PM different under light-load and heavy-load?

3. Type III Compensation:

- a. What are your chosen values of $R_1, R_2, R_3, R_4, C_1, C_2, C_3$ for $PM > 60$ degrees and $GM > 10$ dB (table)
- b. Report the numerical values of GM, PM , and UGB for both light-load and heavy-load conditions (Provide values in table format) and also provide the bode plots with markers at GM and PM crossover frequencies
- c. What is the major benefit of Type III compensation over Type I?

4. Integrate the closed-loop control (Feedback Resistor Divider + ideal OPAMP/OTA with compensation + PWM modulator: ideal comparator with ramp voltage input and SR latch with clock input) to the ideal buck converter designed in Part 1 A. Use ideal components from /analoglib as required.

- a. Perform transient simulation of the closed loop buck without compensation. Does V_{out} settle to desired value? Is the system stable?

*(Remember that small signal AC analysis to determine stability **cannot** be performed with the nonlinear switching circuit, hence we will look at transient response on step load change, refer to the hints section for implementation)*

- b. Integrate Type-I compensation and provide a step-up load change from 100mA to 800mA in 10 ns (rising edge time) and step-down load change to return to 100mA load current in 10 ns (falling edge time). Report the maximum output voltage undershoot, recovery time (90% of baseline) and maximum overshoot and corresponding recovery time in tabular format.
- c. Integrate Type-III compensation and provide a step-up load change from 100mA to 800mA in 10 ns (step time) and step-down load change to return to 100mA load current in 10 ns (step time). Report the maximum output voltage undershoot, recovery time (90% of baseline) and maximum overshoot and corresponding recovery time in tabular format. Compare the transient performance of Type-I vs. Type-III.

Part 2: Transistor level design of supporting circuit block

Due on March 08, 2026, 11:59 PM

Digital submission is preferred over scanned handwritten submission

Transistor level design of PWM Modulator (with comparator and ramp/clock generator) + driver stage (with break before make non-overlapping switch control) + MOSFET switching stage + any other logic circuits (latches/gates)

Show the plots showing the working of each circuit block with the suitable testbench.

Part 2 A : Comparator design + SR Latch

Design and test comparator separately with $t_{\text{delay}} < 10\text{ns}$ (1% of T_{sw}) (you can use a ramp voltage as input to comparator against a reference voltage for testing).

Part 2 B : RMP + CLK circuit

Design the ramp/clock generator (Recall that you will require comparators within the ramp generator circuit as well).

Part 2 C: Integration of RMP + CLK circuit, Comparator design + SR Latch (Part 2A+2B+2C = 10 pts)

Integrate them to complete the PWM modulator.

Apply a single DC voltage (emulating output of error amplifier) to the comparator which is varied between $0.1 V_H$ and $0.9 V$ and show change in duty cycle.

Part 2 D: Buffer + Break before make driver. (5 pts)

Design the Buffer + Break before make driver separately and integrate the PWM modulator designed in Part 2 C, buffer + break before make driver. Consider two capacitors each of 100pF for emulating the switching stage.

Show the plots showing the working with the suitable testbench.

Part 2 E: Replace ideal Opamp and compensation in Part 1 B. (PWM generator and power stage are ideal) (10 pts)

Parallely, design error amplifier (feel free to borrow from previous designs) with $> 40\text{ dB}$ gain and $\sim 100\text{ kHz}$ UGB and integrate with Part 1 B circuit (replacing the ideal 'vcvs' element), close the loop via feedback network, and check load-transient response (only way to test stability!), adjust compensation if required and report the change in values of resistors and capacitors.

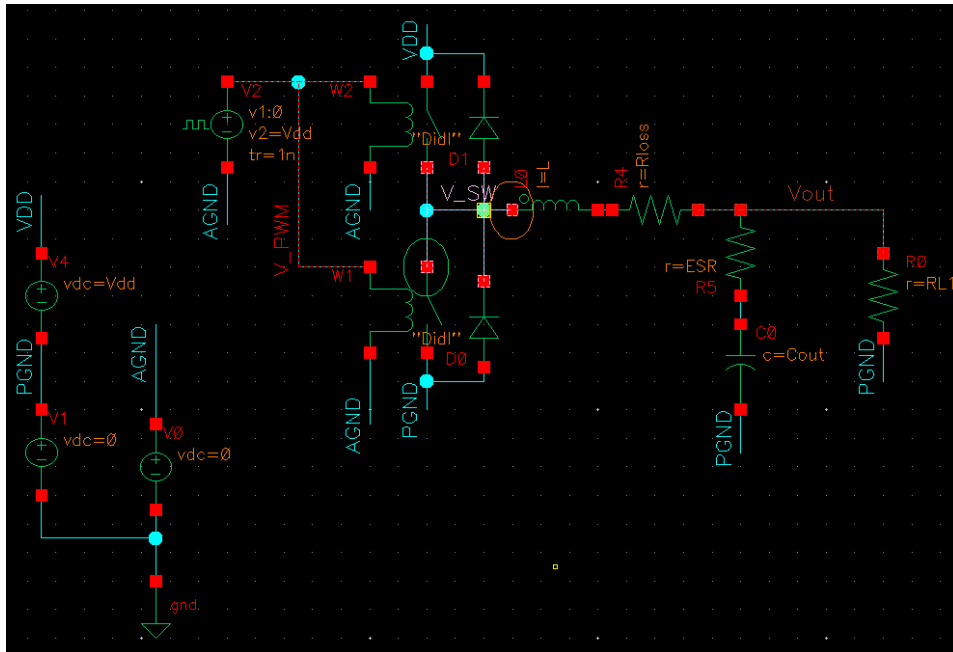
Report the numerical values of GM, PM, and UGB for both light-load and heavy-load conditions (Provide values in table format) and also provide the bode plots with markers at GM and PM crossover frequencies

Part 3: Integration and optimization (20 pts)

Integrate all elements together to complete your design and tune the parameters to meet the specifications!

Appendix:

Power stage design using ideal components

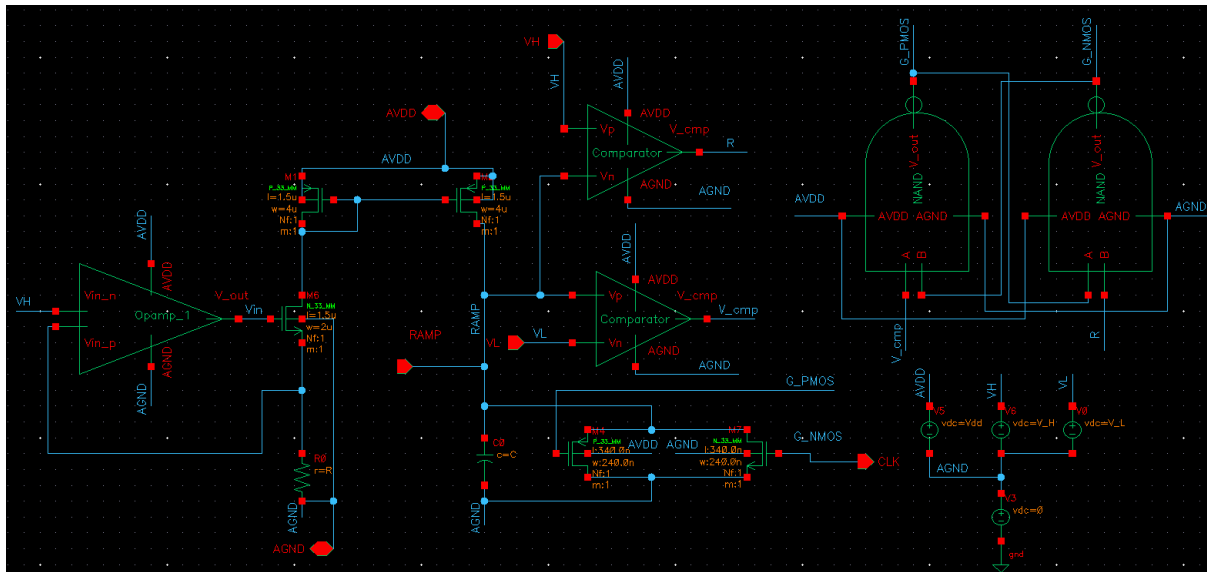


Use analoglib/switch for ideal switch, edit its property and set open and close voltage as 0 and Vdd, respectively for the high-side switch and vice-versa for the low-side switch (this effectively ensures that inverted PWM is provided to the low-side switch). Use analoglib/Vpulse for PWM signal generation, set Period as T (1u) and Pulse width as D*T, you can set the value of variable D from ADE L. Keep rise and fall time to small finite value, say 1ns.

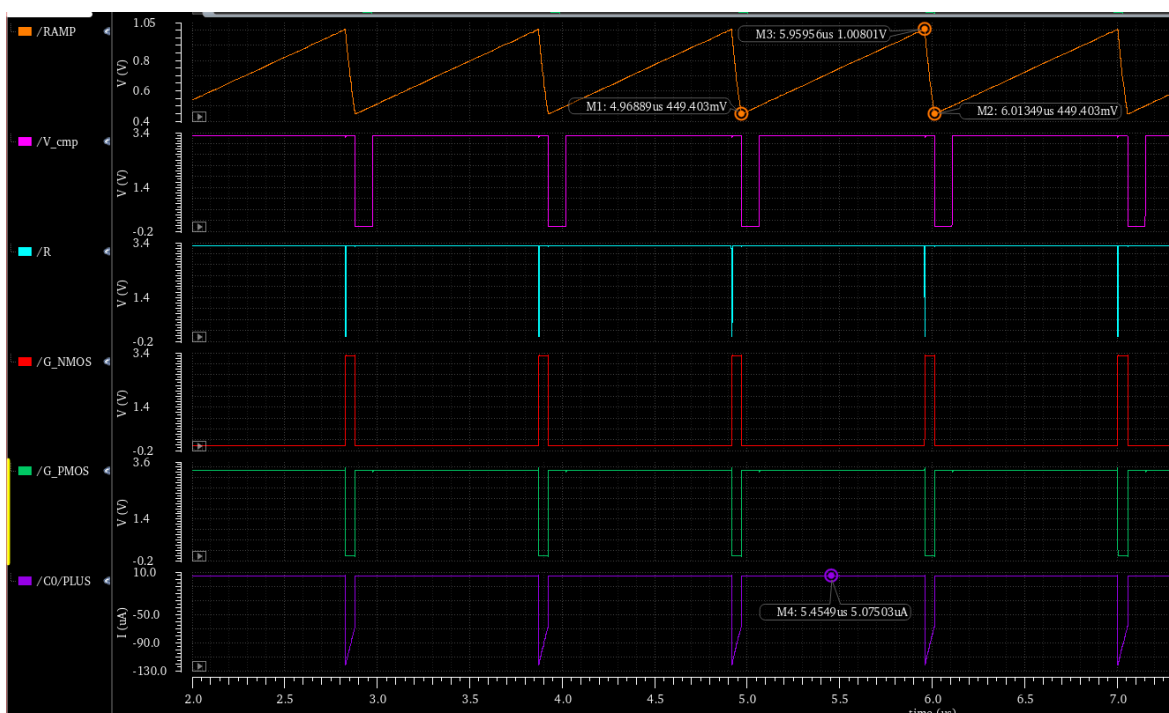
It is important to include body diodes of the switches in simulation, we will use near-ideal diodes for the same. Place analoglib/diode across the two switches. Edit properties of diode and write "Model" as Didl. Then go to the folder containing your project (one with cds.lib file), create a new text file named "Didl.lib" and paste the following: **.model Didl D(Ron=1E-6 Roff=10G Vfwd=0 epsilon=1m revepsilon=1m)**. Next, go to ADE L -> Setup -> Model Libraries and add the path to the created Didl.lib file. You are ready to run the simulation.

Ramp and clock generator design

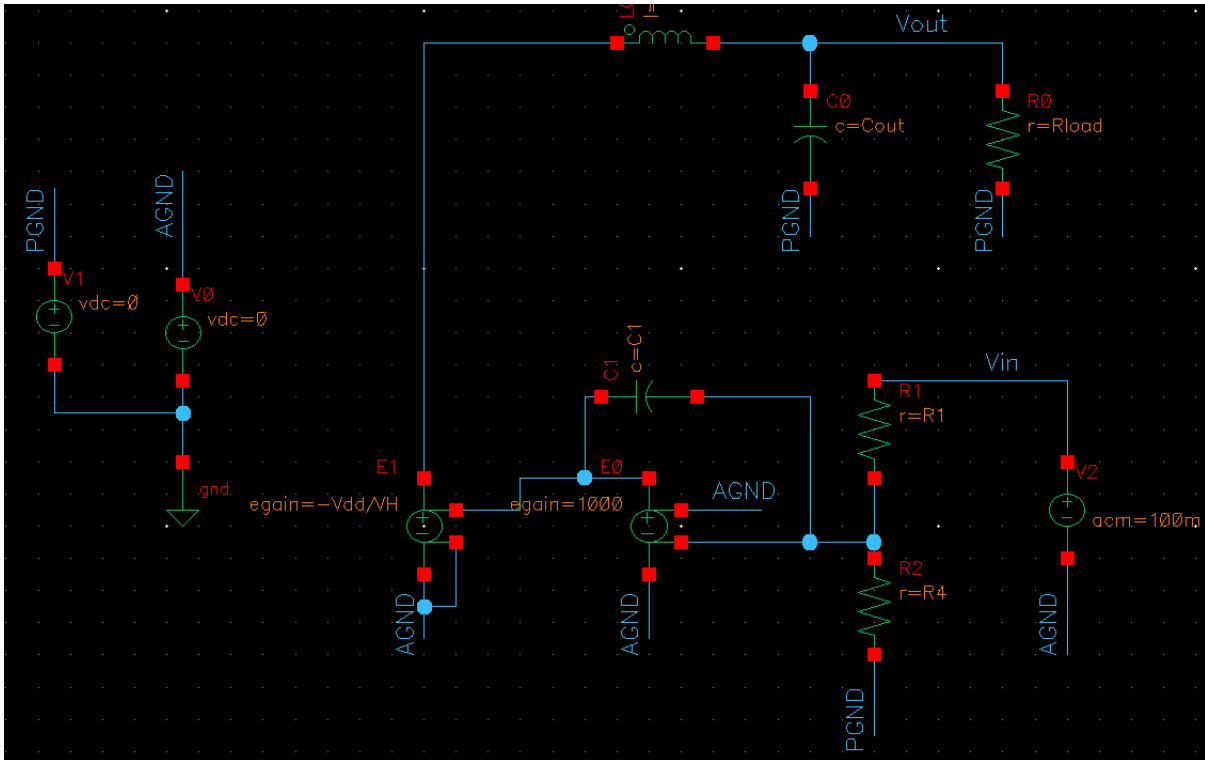
Architecture & Working:



A current mirror is used to set a fixed charging current for capacitor C. When the cap voltage reaches the set VH, comparator with reference voltage VH (High Comparator) gets triggered and issues a RESET pulse SR latch to activate the transmission gate (N-MOS and P-MOS in parallel). This shorts the two ends of capacitor, allowing it to quickly discharge. When the cap voltage reaches lower limit VL, Low Comparator with VL as reference voltage is triggered which deactivates the transmission gate. This completes one cycle and cap charging resumes for the next cycle. It is worth noting that a NMOS and PMOS are used in parallel as a switch instead of a single transistor to enable a wide range of VH and VL values to be compatible with the design. Note that OPAMP in the above circuit is used to generate a current source/bias current, however, you don't require OPAMP if you already have a supply-independent current reference on-board.



Small signal compensator design



Option 1: To emulate ideal Opamp, use analoglib/VCVS with high gain (of the order of 1000 equivalent to 60dB). You may set maximum and minimum voltage of analoglib/VCVS to a reasonable value say +/- Vdd to ensure output remains bound.

Option 2: To emulate ideal OTA, use analoglib/VCCS with a resistor at the output, set gm x resistor = 1000 or higher.

For gain block V_{dd}/V_H , again analoglib/VCVS can be used.

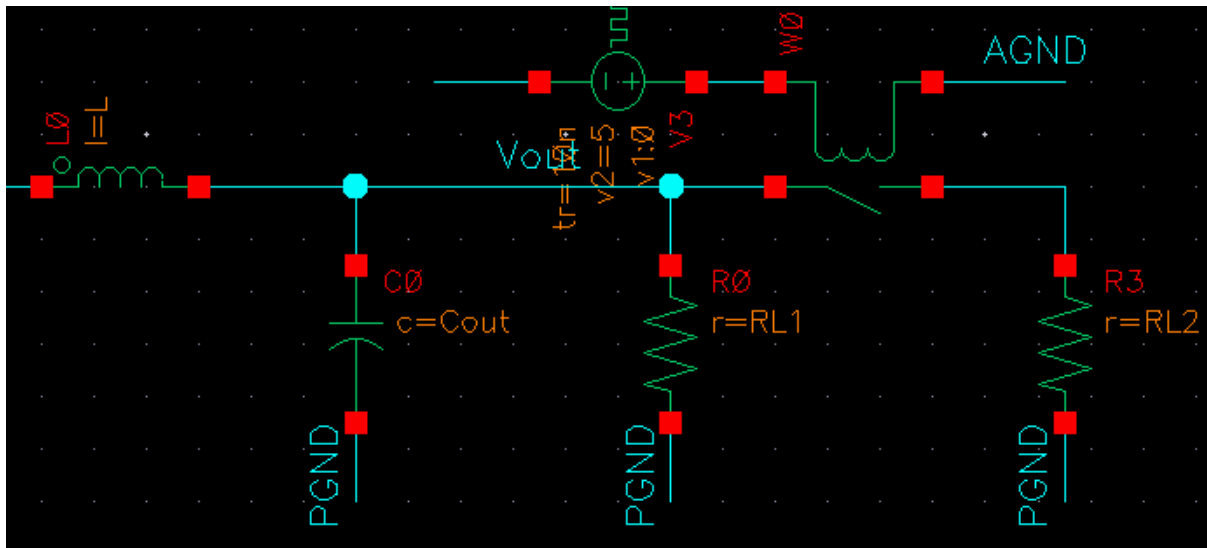
[Demystifying Type II and Type III Compensators Using Op](#) Texas Instrurment Guidelines on Compensation

Some useful Cadence Calculator Functions:

```
dB20((VF("/Vout") / VF("/Vin")))
phase((VF("/Vout") / VF("/Vin")))
phaseMargin((VF("/Vout") / VF("/Vin")))
unityGainFreq((VF("/Vout") / VF("/Vin")))
gainMargin((VF("/Vout") / VF("/Vin")))
```

For load transient simulation:

Place two resistors in parallel as load, with an ideal switch in-between, such that when switch is off R1 is actively emulating light-load condition. When the switch is turned on (programmed to turn-on, for example, after 100us for a period of 100us and then again switch off), R1 and R2 are in parallel, and their equivalent resistance can emulate heavy-load condition. So, total 300us transient simulation. Use analoglib/switch, make sure to edit its properties and make open and closed voltage, for example, 0 and Vdd. Use analoglib/vpulse to control the switch with V1 as 0V and V2 as Vdd, with appropriate rise time and period.



Break before make:

Break before make circuit schematic (courtesy Prof. Cheng Huang, Iowa State University). Note that the final recommended design is a combination of both the break before make techniques.

Buffer Design – Shoot-Through

Combination

